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Sheet discharger, and sheet post-processor, image forming apparatus, and image forming system provided with the sheet discharger

Abstract

A sheet discharger includes a sheet discharge outlet, a discharge member, a sheet discharge tray, a sheet hold-down member, a drive portion, and a control portion. The sheet discharge tray is disposed downstream from the sheet discharge outlet with respect to a discharge direction and is a tray on which a sheet discharged through the sheet discharge outlet is loaded. The sheet hold-down member is swingable between a hold-down position at which the sheet loaded on the sheet discharge tray is held down at an upstream part thereof in the discharge direction and a retracted position retracted upward from the hold-down position. The drive portion drives the sheet hold-down member. The control portion determines, in accordance with characteristics of the sheet, whether or not to execute a sheet hold-down operation in which the sheet hold-down member is moved to the retracted position and to the hold-down position.

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2023-12101 (filed on Jan. 30, 2023), the entire contents of which are incorporated herein by reference.

BACKGROUND

(2) The present disclosure relates to a sheet discharger that discharges a sheet after being subjected to image formation by an image forming apparatus, and a sheet prost-processor, the image forming apparatus, and an image forming system that are provided with the sheet discharger.

(3) An image forming apparatus such as a copy machine or a printer and a sheet post-processor that performs post-processing on a sheet are each provided with a sheet discharger including a discharge roller pair and a discharge tray. A sheet (a sheet of paper) after being subjected to image formation by the image forming apparatus is discharged on the discharge tray by the sheet discharger provided in the image forming apparatus. Alternatively, after being carried into the sheet post-processor and subjected to a prescribed type of post-processing therein, the sheet is discharged on the discharge tray by the sheet discharger provided in the sheet post-processor.

(4) In the sheet discharger described above, in a case where sheets have already been loaded on the discharge tray, a succeeding sheet might push out any of the loaded sheets, causing a loaded state of the sheets to be disordered or the any of the loaded sheets to fall from the discharge tray. Furthermore, in a case of discharging a largely curled sheet, when discharged in such a curled state, the sheet might clog a sheet discharge outlet to cause a sheet discharge failure. As a solution thereto, a method has been adopted in which a sheet discharged on the discharge tray through the sheet discharge outlet is held down at a trailing end thereof.

SUMMARY

(5) A sheet discharger according to an aspect of the present disclosure includes a sheet discharge outlet, a discharge member, a sheet discharge tray, a sheet hold-down member, a drive portion, and a control portion. The sheet discharge outlet is a port through which a sheet is discharged. The discharge member is disposed at the sheet discharge outlet and conveys the sheet in a discharge direction. The sheet discharge tray is disposed downstream from the sheet discharge outlet with respect to the discharge direction and is a tray on which the sheet discharged through the sheet discharge outlet is loaded. The sheet hold-down member is disposed below the sheet discharge outlet and is swingable between a hold-down position at which the sheet loaded on the sheet discharge tray is held down at an upstream part thereof in the discharge direction and a retracted position retracted upward from the hold-down position. The drive portion drives the sheet hold-down member. The control portion controls the drive portion. The control portion is capable of executing, every time the sheet is discharged on the sheet discharge tray, a sheet hold-down operation in which the sheet hold-down member disposed at the hold-down position is moved to the retracted position at a timing when a trailing end of the sheet discharged through the sheet discharge outlet passes through the discharge member, and the sheet hold-down member is moved from the retracted position to the hold-down position at a timing when loading of the sheet on the sheet discharge tray is completed. The control portion determines, in accordance with characteristics of the sheet, whether or not to execute the sheet hold-down operation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 a schematic view showing a configuration of an image forming system composed of a sheet post-processor according to one embodiment of the present disclosure and an image forming apparatus to which the sheet post-processor is connected.
- (2) FIG. 2 is a sectional side view schematically showing a configuration of the sheet post-processor of the present embodiment.
- (3) FIG. 3 is a perspective view of a sheet binding unit mounted in the sheet post-processor of the present embodiment.
- (4) FIG. 4 is a side view of the sheet binding unit.
- (5) FIG. 5 is a sectional side view showing a structure in a vicinity of a processing tray of the sheet binding unit.
- (6) FIG. 6 is a sectional side view schematically showing a vicinity of a first sheet discharge portion of the sheet post-processor, which illustrates a state where sheets have been discharged on a

first discharge tray.

(7) FIG. 7 is a sectional side view schematically showing the vicinity of the first sheet discharge portion of the sheet post-processor, which illustrates a state where downwardly curled sheets have been discharged on the first discharge tray.

(8) FIG. 8 is a flow chart showing an example of operational control of a sheet hold-down member at a time of sheet discharge in the first sheet discharge portion.

DETAILED DESCRIPTION

(9) [1. Configuration of Image Forming System]

(10) With reference to the appended drawings, the following describes an embodiment of the present disclosure in detail. FIG. 1 is a schematic view showing a configuration of an image forming system composed of a sheet post-processor 1 according to one embodiment of the present disclosure and an image forming apparatus 200 to which the sheet post-processor 1 is connected.

(11) As shown in FIG. 1, based on image data externally inputted via an unshown network communication portion or image data read by an image reading portion 201 disposed in an upper part of the image forming apparatus 200, the image forming apparatus 200 prints an image on a sheet (a sheet of paper). In the present embodiment, the image forming apparatus 200 is an inkjet recording apparatus including recording heads (not shown) of different colors each having a multitude of nozzle orifices for ejecting ink to the sheet.

(12) An operation panel 202 is disposed in front of the image reading portion 201. The operation panel 202 is an operation portion for accepting inputs of various types of settings. For example, by operating the operation panel 202, a user can input size information of the sheet. Furthermore, by operating the operation panel 202, a user can also input the number of sheets to be subjected to printing or an instruction to start a printing job. A main body control portion 203 performs centralized control of operations of the image forming apparatus 200 as a whole and controls the various portions of the image forming apparatus 200.

(13) The sheet post-processor 1 is detachably connected to a side surface of the image forming apparatus 200. The sheet post-processor 1 performs post-processing such as a punch hole forming process or a binding process with respect to the sheet after being subjected to image formation (printing) by the image forming apparatus 200. The sheet post-processor 1 is not limited to a configuration of performing post-processing on the sheet automatically conveyed from the image forming apparatus 200 but may also perform post-processing on the sheet placed on an unshown tray by a user and conveyed by the sheet post-processor 1 itself to a position allowing the post-processing to be performed.

(14) [2. Configuration of Sheet Post-Processor]

(15) FIG. 2 is a sectional side view schematically showing a configuration of the sheet post-processor 1 of the present embodiment. As shown in FIG. 2, the sheet post-processor 1 includes a sheet carry-in inlet 2, a first sheet conveyance path 3, a first sheet discharge portion 4, a second sheet conveyance path 5, a second sheet discharge portion 6, a third sheet conveyance path 7, a third sheet discharge portion 8, a post-processing section 9, and a post-processing control portion (a control portion) 10.

(16) The sheet carry-in inlet 2 is an opening provided in a side surface of the sheet post-processor 1 opposed to the image forming apparatus 200. The sheet being conveyed from the image forming apparatus 200 toward the sheet post-processor 1 passes through the sheet carry-in inlet 2 to be carried into the sheet post-processor 1.

(17) The first sheet conveyance path 3 extends substantially horizontally from the sheet carry-in inlet 2 to the first sheet discharge portion 4 in a direction away from the image forming apparatus 200 (a leftward direction in FIG. 2). A direction directed from the sheet carry-in inlet 2 toward the first sheet discharge portion 4 is referred to as a sheet conveyance direction on the first sheet conveyance path 3. The sheet carry-in inlet 2 is positioned at an upstream end of the first sheet conveyance path 3 in the sheet conveyance direction. The first sheet conveyance path 3 includes a

plurality of conveyance roller pairs **3r** and conveys, toward downstream in the sheet conveyance direction, the sheet carried into the sheet post-processor **1** through the sheet carry-in inlet **2**.

(18) The first sheet discharge portion **4** is provided on a side surface of the sheet post-processor **1** opposite to the side surface thereof opposed to the image forming apparatus **200**.

(19) The first sheet discharge portion **4** is disposed at a downstream end of the first sheet conveyance path **3** in the sheet conveyance direction. The first sheet discharge portion **4** includes a first discharge outlet **41**, a first discharge roller pair **42**, and a first discharge tray **43**.

(20) The first discharge outlet **41** is positioned at the downstream end of the first sheet conveyance path **3** in the sheet conveyance direction. The first discharge roller pair **42** is disposed at the first discharge outlet **41**. The first discharge tray **43** is positioned downstream from the first discharge outlet **41** in the sheet conveyance direction. The sheet that has been conveyed on the first sheet conveyance path **3** to reach the first discharge outlet **41** is caused to pass through the first discharge outlet **41** by the first discharge roller pair **42** to be discharged on the first discharge tray **43**. The first discharge tray **43** is one of locations at which the sheet subjected to post-processing by the sheet post-processor **1** is eventually discharged.

(21) The second sheet conveyance path **5** branches off from a first branch portion (a branch portion) **31** on the first sheet conveyance path **3** to extend laterally in the direction away from the image forming apparatus **200** (the leftward direction in FIG. 2) and upwardly to the second sheet discharge portion **6**. The first branch portion **31** is disposed downstream from a perforation portion **91** with respect to the sheet conveyance direction on the first sheet conveyance path **3**. A direction directed from the first branch portion **31** toward the second sheet discharge portion **6** is referred to as a sheet conveyance direction on the second sheet conveyance path **5**. The first branch portion **31** is positioned at an upstream end of the second sheet conveyance path **5** in the sheet conveyance direction. The second sheet conveyance path **5** includes a plurality of conveyance roller pairs **5r** and conveys the sheet being conveyed on the first sheet conveyance path **3** so that a conveyance direction of the sheet is changed at the first branch portion **31** toward the second sheet discharge portion **6**.

(22) The first branch portion **31** includes a first switching guide **311**. The first switching guide **311** pivots to a position at which the sheet being conveyed on the first sheet conveyance path **3** from the sheet carry-in inlet **2** is guided to the first discharge outlet **41** along the first sheet conveyance path **3** and to a position at which the conveyance direction of the sheet is changed from that on the first sheet conveyance path **3** so that the sheet is guided to the second sheet conveyance path **5**. The first switching guide **311** further pivots to a position at which the sheet that has been subjected to a folding process is guided to the second sheet conveyance path **5**. The first switching guide **311** is connected to a drive mechanism (not shown), and an operation thereof is controlled by a post-processing control portion **10**.

(23) The second sheet discharge portion **6** is provided on the side surface of the sheet post-processor **1** opposite to the side surface thereof opposed to the image forming apparatus **200** and above the first sheet discharge portion **4**. The second sheet discharge portion **6** is disposed at a downstream end of the second sheet conveyance path **5** in the sheet conveyance direction. The second sheet discharge portion **6** includes a second discharge outlet **61**, a second discharge roller pair **62**, and a second discharge tray **63**.

(24) The second discharge outlet **61** is positioned at the downstream end of the second sheet conveyance path **5** in the sheet conveyance direction. The second discharge roller pair **62** is disposed at the second discharge outlet **61**. The second discharge tray **63** is positioned downstream from the second discharge outlet **61** in the sheet conveyance direction. The sheet that has been conveyed on the second sheet conveyance path **5** to reach the second discharge outlet **61** is caused to pass through the second discharge outlet **61** by the second discharge roller pair **62** to be discharged on the second discharge tray **63**. The second discharge tray **63** is one of the locations at which the sheet subjected to post-processing by the sheet post-processor **1** is eventually discharged.

Furthermore, a sheet not to be subjected to post-processing, a small-size sheet, and so on are also discharged on the second discharge tray **63**.

(25) The third sheet conveyance path **7** branches off from a second branch portion **32** on the first sheet conveyance path **3** to extend downwardly to the third sheet discharge portion **8**. A direction directed from the second branch portion **32** toward the third sheet discharge portion **8** is referred to as a sheet conveyance direction on the third sheet conveyance path **7**. The second branch portion **32** is positioned downstream from the first branch portion **31** with respect to the sheet conveyance direction on the first sheet conveyance path **3** and at an upstream end of the third sheet conveyance path **7** in the sheet conveyance direction. The third sheet conveyance path **7** includes a plurality of conveyance roller pairs **7r** and conveys the sheet being conveyed on the first sheet conveyance path **3** so that the conveyance direction of the sheet is changed at the second branch portion **32** toward the third sheet discharge portion **8**.

(26) The second branch portion **32** includes a second switching guide **321**. The second switching guide **321** pivots to a position at which the sheet being conveyed on the first sheet conveyance path **3** from the sheet carry-in inlet **2** is guided to the first discharge outlet **41** along the first sheet conveyance path **3** and to a position at which the sheet that has been conveyed on the first sheet conveyance path **3** from the sheet carry-in inlet **2** to pass through the second branch portion **32** and has then been switched back is guided to the third sheet conveyance path **7**. The second switching guide **321** is connected to a drive mechanism (not shown), and an operation thereof is controlled by the post-processing control portion **10**.

(27) The third sheet discharge portion **8** is provided on the side surface of the sheet post-processor **1** opposite to the side surface thereof opposed to the image forming apparatus **200** and below the first sheet discharge portion **4** (near a lower end of the sheet post-processor **1**). The third sheet discharge portion **8** includes a third discharge outlet **81**, a third discharge roller pair **82**, and a third discharge tray **83**.

(28) The third discharge outlet **81** is positioned at a downstream end of the third sheet conveyance path **7** in the sheet conveyance direction. The third discharge roller pair **82** is disposed at the third discharge outlet **81**. The third discharge tray **83** is positioned downstream from the third discharge outlet **81** in the sheet conveyance direction. The sheet that has been conveyed on the third sheet conveyance path **7** to reach the third discharge outlet **81** is caused to pass through the third discharge outlet **81** by the third discharge roller pair **82** to be discharged on the third discharge tray **83**. The third discharge tray **83** is one of the locations at which the sheet subjected to post-processing by the sheet post-processor **1** is eventually discharged.

(29) The post-processing section **9** performs a prescribed type of post processing with respect to the sheet that has been subjected to image formation by the image forming apparatus **200** and has been carried into the sheet post-processor **1**. The post-processing section **9** includes the perforation portion **91**, a sheet binding unit **92**, a sheet folding unit **100**, and a bookbinding portion **94**.

(30) The perforation portion **91** is disposed in immediate proximity to and downstream from the sheet carry-in inlet **2** on the first sheet conveyance path **3**. The perforation portion **91** performs a perforation process with respect to the sheet being conveyed on the first sheet conveyance path **3** so as to form a punch hole therein.

(31) The sheet binding unit **92** is disposed in immediate proximity to and upstream from the first sheet discharge portion **4** with respect to the sheet conveyance direction on the first sheet conveyance path **3**. The sheet binding unit **92** performs a staple process (the binding process) with respect to a sheet bundle of a plurality of sheets stacked on each other so as to bind the sheet bundle. A detailed configuration of the sheet binding unit **92** will be described later.

(32) The sheet folding unit **100** is disposed downstream from the perforation portion **91** and upstream from the sheet binding unit **92** with respect to the sheet conveyance direction on the first sheet conveyance path **3**. The sheet folding unit **100** performs the folding process with respect to a single sheet so as to form a fold line on the sheet. The sheet folding unit **100** is capable of

performing, with respect to a single sheet, the folding process for folding the sheet, for example, in two, in a Z shape, outwardly in three, or inwardly in three.

(33) The bookbinding portion **94** is disposed in immediate proximity to and upstream from the third sheet discharge portion **8** with respect to the sheet conveyance direction on the third sheet conveyance path **7**. The bookbinding portion **94** includes a center-folding part **941** and a center-binding part **942**. The bookbinding portion **94** performs, with respect to a sheet bundle of a plurality of sheets stacked on each other, a center-folding process and a center-binding process in which the sheet bundle is folded at substantially a center thereof in the sheet conveyance direction and bound into a booklet.

(34) The post-processing control portion (the control portion) **10** includes a CPU, a storage part, and other electronic circuits and electronic components (none of which are shown). The post-processing control portion **10** is communicably connected to the main body control portion **203** in the image forming apparatus **200** (see FIG. 1). Upon receipt of an instruction from the main body control portion **203**, based on control programs and data stored in the storage part by use of the CPU, the post-processing control portion **10** controls operations of the various constituent elements provided in the sheet post-processor **1** so as to perform processing related to functions of the sheet post-processor **1**. The first sheet conveyance path **3**, the first sheet discharge portion **4**, the second sheet conveyance path **5**, the second sheet discharge portion **6**, the third sheet conveyance path **7**, the third sheet discharge portion **8**, and the post-processing section **9** individually receive instructions from the post-processing control portion **10** and perform post-processing on the sheet in conjunction with each other. A configuration may be adopted in which the main body control portion **203** in the image forming apparatus **200** also assumes functions of the post-processing control portion **10**.

(35) [3. Configuration of Sheet Binding Unit]

(36) Next, a description is given of a configuration of the sheet binding unit **92**. FIG. 3 is a perspective view of the sheet binding unit **92** mounted in the sheet post-processor **1**. FIG. 4 is a side view of the sheet binding unit **92**.

(37) As shown in FIG. 3, the sheet binding unit **92** includes a processing tray **521**, a staple portion **71**, and reference plates **73**.

(38) The processing tray **521** is a rectangular tray extending in a sheet width direction (a direction of arrows A and A') and in a carry-in direction. A plurality of sheets S (a sheet bundle) to be subjected to the staple process is loaded on the processing tray **521**. In this case, the sheets S are carried onto the processing tray **521** along an alignment direction (opposite to the carry-in direction) directed toward a lower rightward direction (an arrow B direction) in FIG. 4. The sheet bundle that has been subjected to the staple process is eventually sent out toward a direction (an upper leftward direction in FIG. 4) opposite to the above-described alignment direction by the first discharge roller pair **42** (see FIG. 2) to be discharged on the first discharge tray **43** (see FIG. 2). Lower discharge rollers **421** each used as one of rollers constituting the first discharge roller pair **42** are supported at a downstream part of the processing tray **521** in the carry-in direction (a lower left side in FIG. 3).

(39) The processing tray **521** includes a tray center part **522** and width regulation members **523**. On an upper surface of the processing tray **521**, the tray center part **522** is disposed at a center in the sheet width direction. The tray center part **522** is a thin plate-shaped member secured onto the processing tray **521** at a slight height therefrom.

(40) A pair of width regulation members **523** is disposed so as to interpose the tray center part **522** therebetween in the sheet width direction. The width regulation members **523** regulate a position of the sheet S carried onto the processing tray **521** in the sheet width direction. Each of the width regulation members **523** is formed of a thin plate-shaped member similarly to the tray center part **522** and has an upwardly standing side wall provided at an end thereof in the sheet width direction. The processing tray **521** has a guide groove **524** formed therein to extend along the sheet width

direction. The width regulation members **523** are reciprocable in the sheet width direction along the guide groove **524** via a drive mechanism (not shown) such as a rack and a pinion gear. In the present embodiment, every time the sheet is carried onto the processing tray **521**, the width regulation members **523** are caused to reciprocate by the drive mechanism. As a result, the sheet loaded on the processing tray **521** is aligned in the sheet width direction.

(41) The staple portion **71** is disposed to be opposed to an end edge of the sheet on a leading end side thereof in the alignment direction (a right side in FIG. **4**). The staple portion **71** is movable in the sheet width direction (the direction of arrows A and A') orthogonal to the carry-in direction along the end edge of the sheet by a drive force of a staple portion drive motor M1 and performs the staple process on the sheet bundle.

(42) As shown in FIG. **4**, the staple portion **71** includes a staple main body part **711** and a staple movable part **712**. The staple main body part **711** is a main body of the staple portion **71** and contains therein a plurality of staples (not shown). The staple movable part **712** is movable up and down and drives a staple into the sheets S. A concave **713** for receiving the end edge of the sheet is formed between the staple main body part **711** and the staple movable part **712**.

(43) The reference plates **73** are secured respectively at three locations spaced from each other in the sheet width direction so as to be opposed to a downstream end of the processing tray **521** in the alignment direction (an upper right side in FIG. **3**, a lower right side in FIG. **4**). The reference plates **73** each have substantially a U shape open toward upstream in the alignment direction (an upper left side in FIG. **4**) when viewed in a cross section orthogonal to the sheet width direction. The reference plates **73** abut on the end edge of the sheet carried onto the processing tray **521**, thus aligning the sheet in the alignment direction.

(44) [4. Configuration in Vicinity of Processing Tray in Sheet Binding Unit]

(45) FIG. **5** is a sectional side view showing a structure in a vicinity of the processing tray **521** and an alignment member **55**. As shown in FIG. **5**, a carry-in roller pair **54** is disposed above the processing tray **521**. The carry-in roller pair **54** is composed of an upper carry-in roller **54a** and a lower carry-in roller **54b**.

(46) A sheet detection sensor **93** is disposed in a neighborhood of the carry-in roller pair **54**. The sheet detection sensor **93** detects a timing when the sheet S passes through the carry-in roller pair **54**. As the sheet detection sensor **93**, there is used, for example, a PI (photointerrupter) sensor including a detection portion composed of a light emitting part and a light receiving part.

(47) A tapping member **53** and the alignment member **55** are provided downstream from the carry-in roller pair **54** with respect to a carry-in direction of the sheet S (a left side in FIG. **5**). The tapping member **53** is supported so as to be swingable along the carry-in direction of the sheet S. At a timing when a trailing end of the sheet S passes through the carry-in roller pair **54**, the tapping member **53** swings downward to tap the sheet S downward so that the sheet S is guided onto the processing tray **521**.

(48) The alignment member **55** is disposed at each of a plurality of locations (four locations in the present embodiment) along the sheet width direction (a direction perpendicular to a plane of FIG. **5**). The alignment member **55** assists in aligning the sheet S carried onto the processing tray **521** by moving (switching back) the sheet S in the alignment direction approaching the reference plates **73**. The alignment member **55** includes a paddle holder **56** and an alignment paddle **57**.

(49) The paddle holder **56** is supported above the processing tray **521** so as to be swingable along the carry-in direction of the sheet S. By a holder drive motor (not shown), a rotation drive force is inputted to a swing shaft **56a** of the paddle holder **56**. By a drive source (not shown) such as a motor, a rotation drive force in such a direction that the sheet S is sent out in the alignment direction (a counterclockwise direction in FIG. **5**) is inputted to the alignment paddle **57**. The alignment paddle **57** rotates while abutting on an upper surface of the sheet S carried onto the processing tray **521** to move the sheet S in the alignment direction so that an end edge of the sheet S strikes the reference plates **73**, thus aligning the sheet S.

(50) Swinging of the paddle holder **56** is controlled based on a timing of detection by the sheet detection sensor **93**. Specifically, at a timing when the sheet detection sensor **93** detects passing of a leading end of the sheet **S** through the carry-in roller pair **54**, the paddle holder **56** is swung upward. As a result, the alignment paddle **57** is separated from the upper surface of the processing tray **521** (or of the sheet **S** loaded on the processing tray **521**).

(51) FIG. **5** shows a state immediately before a new sheet **S** is carried onto the processing tray **521** in which the paddle holder **56** is swung upward (in a clockwise direction) and the alignment paddle **57** is disposed at a position (a reference position) separated from the processing tray **521**. Furthermore, there is released a nip between each of the lower discharge rollers **421** and a corresponding one of upper discharge rollers **422**, which constitute the first discharge roller pair **42**. Thus, the sheet **S** that has been carried from the carry-in roller pair **54** onto the processing tray **521** once passes through the first discharge roller pair **42** to project over the first discharge tray **43**.

(52) Further, at a timing when a training end of the new sheet **S** carried onto the processing tray **521** passes below the alignment paddle **57**, the paddle holder **56** is swung in a reverse direction (the counterclockwise direction). As a result, the alignment paddle **57** is disposed at a position (an acting position) at which the alignment paddle **57** contacts the upper surface of the sheet **S**. The above-described operation is repeatedly performed every time the sheet **S** is carried thereonto, and thus it is possible to achieve reliable contact of the alignment paddle **57** with the upper surface of the sheet **S** while avoiding interference between the leading end of the sheet **S** being carried onto the processing tray **521** and the alignment paddle **57**.

(53) A support member **58** is disposed below the processing tray **521**. The support member **58** is a rod-shaped member having a prescribed width in the sheet width direction and extending in an arc shape in a discharge direction and is disposed below the first discharge outlet **41**. To be more specific, the support member **58** is disposed below the processing tray **521** and below relative to a discharge path of the sheet **S** discharged from the first discharge roller pair **42** along the processing tray **521**. In the present embodiment, the support member **58** is disposed at each of two locations in the sheet width direction spaced at a prescribed distance from each other with respect to a center of the processing tray **521** in the sheet width direction.

(54) By a support member drive motor (not shown), the support member **58** is movable between a projected position (a position indicated by a solid line in FIG. **5**) projected downstream (a left side in FIG. **5**) relative to the first discharge roller pair **42** in the discharge direction and a retracted position (a position indicated by a broken line in FIG. **5**) retracted upstream (a right side in FIG. **5**) relative to the first discharge roller pair **42** in the discharge direction. When the sheet is carried onto the processing tray **521** (when the sheet is switched back), the support member **58** is disposed at the projected position so as to support a part of the sheet projecting over the first discharge tray **43**.

(55) A sheet hold-down member **59** is disposed below the first discharge roller pair **42**. The sheet hold-down member **59** is disposed at each of two locations in the sheet width direction spaced at a prescribed distance from each other with respect to the center in the sheet width direction.

(56) The sheet hold-down member **59** is a rod-shaped member having a prescribed width in the sheet width direction and extending in substantially an up-down direction. The sheet hold-down member **59** is supported at a lower end thereof rotatably about a swing shaft **59a** extending along the sheet width direction as a fulcrum. Based on a control signal from the post-processing control portion **10** (see FIG. **2**), the sheet hold-down member **59** swings about the swing shaft **59a**, with an upper end thereof being a free end. By a hold-down member drive motor **M2** (see FIG. **6**), the sheet hold-down member **59** is displaced between a hold-down position (a position indicated by a broken line in FIG. **5**) at which a bundle of sheets **S** (hereinafter, simply referred to also as the sheets **S**) loaded on the first discharge tray **43** is held down from above at an upstream part thereof in the discharge direction and a retracted position (a position indicated by a solid line in FIG. **5**) at which a held-down state of the sheets **S** is released.

(57) Before a start of an operation of discharging the sheet **S**, the sheet hold-down member **59** is

disposed at the hold-down position at which the sheets S loaded on the first discharge tray **43** are held down from above at upstream parts thereof in the discharge direction. When the sheet S is discharged through the first discharge outlet **41**, at a timing when the trailing end of the sheet S passes through the first discharge roller pair **42**, the sheet hold-down member **59** is moved to the retracted position not projecting toward the first discharge tray **43** (not interfering with discharge of the sheet S). Further, at a timing when loading of the sheet S on the first discharge tray **43** is completed, the sheet hold-down member **59** is moved again from the retracted position to the hold-down position. Every time the sheet S is discharged on the first discharge tray **43**, the sheet hold-down member **59** repeatedly performs a sheet hold-down operation described above (movement between the retracted position and the hold-down position).

(58) [5. Operational Control of Sheet Hold-Down Member]

(59) Next, a description is given of operational control of the sheet hold-down member **59** in the first sheet discharge portion **4**. In the present embodiment, as shown in FIG. **6**, in a case where the sheet S discharged through the first discharge outlet **41** is uncurled or curled upwardly, the sheet hold-down member **59** is moved from the retracted position (the position indicated by the solid line in FIG. **5**) to the hold-down position.

(60) On the other hand, as shown in FIG. **7**, in a case where the sheet S discharged through the first discharge outlet **41** is curled downwardly, the sheet hold-down member **59** is retained at the retracted position without being moved to the hold-down position. That is, the sheets S loaded on the first discharge tray **43** are not held down at the trailing ends thereof by the sheet hold-down member **59**.

(61) Specifically, the post-processing control portion **10** detects a curled state of the sheet S discharged on the first discharge tray **43**. Further, in the case where the sheet S is uncurled or curled upwardly, every time a bundle of sheets S is discharged, there is executed the sheet hold-down operation in which the sheet hold-down member **59** is moved between the retracted position and the hold-down position. On the other hand, in the case where the sheet S is curled downwardly, the sheet hold-down member **59** is not moved from the retracted position to the hold-down position.

(62) According to the above-described configuration, only in the case where the sheet S is uncurled or curled upwardly, the sheet hold-down member **59** is moved to the hold-down position, while in the case where the sheet S is curled downwardly, the sheet hold-down member **59** is not moved from the retracted position, and thus it is possible to suppress a phenomenon in which downwardly curled sheets S on the first discharge tray **43** bulge at centers thereof when held down at trailing ends thereof. Accordingly, when discharging the sheet S on the first discharge tray **43**, it is possible to suppress interference with the sheets S that have already been loaded on the first discharge tray **43** and thus to suppress defects such as curling of the sheet S at the leading end thereof and positional shifting of the sheet.

(63) A direction of curl of the sheet S varies depending on an orientation of a printing surface of the sheet S. For example, in a case where a sheet of regular paper is used as the sheet S, when ink adheres to the sheet S as a result of printing thereon in the image forming apparatus **200**, the printing surface of the sheet S swells with the ink and thus is deformed into a convex shape. That is, the sheet S is curled downwardly when discharged with the printing surface facing upward and is curled upwardly when discharged with the printing surface facing downward.

(64) In a case, however, where the image forming apparatus **200** is an electrophotographic apparatus such as a laser printer, the printing surface (a toner adhering surface) of the sheet S is heated by a heating member (a fixing roller, a fixing belt, or the like) in a fixing portion and thus shrinks to be deformed into a concave shape. That is, the sheet S is curled upwardly when discharged with the printing surface facing upward and is curled downwardly when discharged with the printing surface facing downward.

(65) Furthermore, when the sheet S is a sheet of special paper different from regular paper, the sheet S may be deformed oppositely to the above.

(66) Furthermore, the direction and an amount of curl of the sheet S vary depending also on a type (a thickness, a basis weight, a grain direction) of the sheet S. There is a tendency that the larger the thickness and basis weight of the sheet S, the less likely the leading end of the sheet S is curled, and the smaller the thickness and basis weight of the sheet S, the more likely the leading end of the sheet S is curled.

(67) When the sheet S is a sheet of paper, fibers forming the sheet S tend to be arranged along a direction in which paper travels in a paper machine. The direction in which the fibers are arranged is referred to as the “grain direction” of the sheet S. In a case where the grain direction of the sheet S is parallel to the width direction (a short grain), the sheet S tends to be more likely to be curled. On the other hand, in a case where the grain direction of the sheet S is parallel to the conveyance direction (a long grain), the sheet S tends to be less likely to be curled.

(68) Furthermore, in a case where the image forming apparatus **200** connected to the sheet post-processor **1** is an inkjet recording apparatus, there is a tendency that the larger an amount of ink used for image recording, the larger a moisture amount of the sheet S, and the more likely the sheet S is curled.

(69) In view of the above, in the present embodiment, based on output information related to characteristics of the sheet S, the direction of curl of the sheet S is estimated. Specifically, relationships between the output information related to characteristics of the sheet S (the type of the sheet S such as the thickness, basis weight, grain direction, and so on of the sheet S, the orientation of the printing surface thereof, and the amount of ink thereon) and the direction and amount of curl thereof are experimentally determined in advance. Further, those relationships are tabulated and stored in a storage region (a memory) of the main body control portion **203**. Based on an inputted piece of output information related to characteristics of the sheet S, the post-processing control portion **10** estimates the direction of curl of the sheet S. Further, based on a result of the estimation, it is determined whether or not to move the sheet hold-down member **59** from the retracted position to the hold-down position.

(70) FIG. **8** is a flow chart showing an example of operational control of the sheet hold-down member **59** at a time of sheet discharge in the first sheet discharge portion **4**, which is executed in the sheet post-processor **1** of the present embodiment. With reference to FIG. **1** to FIG. **7** where necessary, a description is given of a procedure of an operation of the sheet hold-down member **59** by following steps shown in FIG. **8**. It is assumed that, in an initial state, the sheet hold-down member **59** is disposed at the retracted position (the position indicated by the solid line in FIG. **5**). Furthermore, in the following description, the output information related to characteristics of the sheet S is simply referred to also as sheet output information.

(71) When a binding process command to perform the binding process on the sheet S is inputted from the main body control portion **203** in the image forming apparatus **200**, the output information on the sheet S is inputted together with the binding process command (step **S1**). Among pieces of the output information on the sheet S, pieces of information related to the type of the sheet S such as a size, the thickness, basis weight, grain direction, and so on of the sheet S are inputted via the operation panel **202** of the image forming apparatus **200**. For example, pieces of sheet information such as a name of a manufacturer, a product name, a product number, and so on of the sheet S may be stored in advance in the post-processing control portion **10** (or the main body control portion **203**) so as to be associated with the direction of curl of the sheet S. Thus, at a start of a printing job, the name of a manufacturer, product name, product number, and so on of the sheet S used are simply selected via the operation panel **202**, and this enables the post-processing control portion **10** to recognize pieces of information related to the type of the sheet S used in the printing job. An input of the orientation of the printing surface is made via the operation panel **202** of the image forming apparatus **200**.

(72) As for a piece of information related to the amount of ink, based on image data transmitted from a host apparatus such as a personal computer or the image reading portion **201**, an amount of

ink used for image recording is calculated by the main body control portion **203** and transmitted from the main body control portion **203**.

(73) Next, the post-processing control portion **10** starts to carry, onto the processing tray **521**, the sheet **S** that has been carried into the sheet post-processor **1** via the sheet carry-in inlet **2** (step **S2**). Specifically, upon the sheet detection sensor **93** detecting that the trailing end of the sheet **S** has passed through the carry-in roller pair **54**, the post-processing control portion **10** (see FIG. **2**) controls the tapping member **53** to tap the trailing end of the sheet **S** so that the sheet **S** is guided onto the processing tray **521** and then controls the paddle holder **56** to swing downward for a prescribed amount. This causes the alignment paddle **57** to move to the acting position at which a paddle part **572** thereof contacts the upper surface of the sheet **S**. The alignment paddle **57** is rotated in this state, thus causing the sheet **S** to be drawn in the alignment direction (the arrow **B** direction) along the processing tray **521**.

(74) After that, the sheet **S** is sent further downstream in the alignment direction by the alignment paddle **57** and loaded in a state of being aligned in the sheet width direction by the width regulation members **523** (see FIG. **3**) and aligned in the alignment direction by the reference plates **73** (see FIG. **3**).

(75) The post-processing control portion **10** counts the number of sheets **S** carried onto the processing tray **521** (step **S3**). Further, every time a single (or a prescribed number of) sheet(s) **S** is/are carried thereonto, a stop position of the paddle holder **56** is moved upward so that the acting position of the alignment paddle **57** is moved upward in a continuous manner.

(76) Next, the post-processing control portion **10** determines whether or not the prescribed number of sheets **S** have been carried onto the processing tray **521** (step **S4**). In a case where the prescribed number of sheets **S** have not been carried thereonto (No at step **S4**), a return is made to step **S3** at which carrying of the sheet **S** onto the processing tray **521** and counting of the number of sheets **S** carried thereonto are continuously performed.

(77) In a case where the prescribed number of sheets **S** have been carried thereonto (Yes at step **S4**), the post-processing control portion **10** transmits a control signal to the staple portion drive motor **M1** (see FIG. **3**) so as to move the staple portion **71** to a prescribed staple position. After the staple portion **71** has moved to the staple position, the post-processing control portion **10** transmits a control signal to the staple portion **71** so as to execute the staple process on the plurality of sheets **S** aligned by the reference plates **73** (step **S5**).

(78) Next, the post-processing control portion **10** controls the rollers constituting the first discharge roller pair **42** to contact each other (form a nip) and the first discharge roller pair **42** to rotate in the discharge direction. This causes the bundle of sheets **S** thus subjected to the staple process to be discharged on the first discharge tray **43** (both of these are shown in FIG. **2**) (step **S6**).

(79) Based on an inputted piece of output information on the sheet **S**, the post-processing control portion **10** determines whether or not the sheet hold-down member **59** needs to be moved from the retracted position to the hold-down position (step **S7**). In a case where it is determined that the sheet hold-down member **59** needs to be moved to the hold-down position (Yes at step **S7**), the post-processing control portion **10** transmits a control signal to the hold-down member drive motor **M2** (see FIG. **6**) so that every time a single bundle of sheets **S** is discharged on the first discharge tray **43**, movement of the sheet hold-down member **59** from the retracted position to the hold-down position is repeatedly executed (step **S8**).

(80) For example, in a case where, based on the piece of output information on the sheet **S** (the thickness, basis weight, and grain direction of the sheet **S**, the orientation of the printing surface thereof, the amount of ink thereon), the sheet **S** is predicted to be uncurled or curled upwardly, it is determined that the sheet hold-down member **59** needs to be moved from the retracted position to the hold-down position.

(81) On the other hand, in a case where it is determined that the sheet hold-down member **59** does not need to be moved to the hold-down position (No at step **S7**), the post-processing control portion

10 performs control so that the sheets **S** are discharged on the first discharge tray **43** without the sheet hold-down member **59** being moved from the retracted position. For example, in a case where, based on the piece of output information on the sheet **S**, the sheet **S** is predicted to be curled downwardly, it is determined that the sheet hold-down member **59** does not need to be moved to the hold-down position.

(82) After the bundle of sheets **S** has been discharged, the post-processing control portion **10** transmits a control signal to the staple portion drive motor **M1** so as to move the staple portion **71** to a standby position (step **S9**).

(83) After that, the post-processing control portion **10** determines whether or not the binding process has ended (step **S10**). In a case where the binding process is being continuously performed (No at step **S10**), a return is made to step **S2**, and there are repeatedly performed carrying of the sheet **S** onto the processing tray **521** and counting of the number of sheets **S** carried thereonto, execution of the staple process by the staple portion **71**, determination as to whether or not the sheet hold-down member **59** needs to be moved, discharging of a sheet bundle, and movement of the staple portion **71** to the standby position (steps **S2** to **S9**). In a case where the binding process has ended (Yes at step **S10**), the process is ended without any further action.

(84) According to the control example shown in FIG. **8**, based on the output information on the sheet **S**, it is determined whether or not the sheet hold-down operation by the sheet hold-down member **59** needs to be performed. Therefore, the sheet **S** can be stably discharged on the first discharge tray **43** and loaded thereon in an aligned state without being affected by a direction of curl of the sheet **S** varying depending on the type of the sheet **S** and the amount of ink used for image recording by an inkjet recording apparatus.

(85) While the foregoing has described the embodiment of the present disclosure, the present disclosure is not limited in scope thereto and can be implemented by being variously modified without departing from the spirit of the invention. For example, while in the foregoing embodiment, based on the output information on the sheet **S**, the post-processing control portion **10** automatically determines whether or not to execute the sheet hold-down operation by the sheet hold-down member **59**, a configuration may be adopted in which a user can move the sheet hold-down member **59** at any timing. For example, in a possible configuration, the operation panel **202** has a maintenance mode in which a position of the sheet hold-down member **59** is switched between the retracted position and the hold-down position so that a user can switch the position of the sheet hold-down member **59** in accordance with a loaded status (an alignment property) of the sheets **S** on the first discharge tray **43**.

(86) With the above-described configuration, even in a case where a degree of downward curl of the sheet **S** is actually larger than assumed, it is possible to stop the sheet hold-down member **59** from moving to the hold-down position in consideration of a discharged status of a bundle of sheets **S** on the first discharge tray **43**.

(87) Furthermore, while the foregoing embodiment describes an example in which, by the sheet binding unit **92**, the binding process is performed on the sheets **S** loaded on the processing tray **521**, the present disclosure is not limited thereto, and a configuration may be adopted in which a shift discharge process or the folding process is performed on the sheets **S** loaded on the processing tray **521**.

(88) Furthermore, while in the foregoing embodiment, information related to the type of the sheet **S** is inputted via the operation panel **202** of the image forming apparatus **200**, a configuration may be adopted in which the information related to the type of the sheet **S** is automatically acquired. For example, a media sensor can be disposed at any position on a sheet conveyance path from the image forming apparatus **200** to the sheet post-processor **1** and used to detect information related to the sheet **S** such as the thickness, basis weight, and grain direction of the sheet **S** carried from the image forming apparatus **200** into the sheet post-processor **1** and the amount of ink thereon.

(89) For example, in detecting the thickness of the sheet **S**, there can be used, as the media sensor, a

laser coaxial displacement meter that includes two optical sensors used to detect the thickness of the sheet S by interposing the sheet S therebetween.

(90) In detecting the basis weight of the sheet S, there can be used, as the media sensor, a basis weight sensor that uses a light transmittance through the sheet S to measure the basis weight. A relationship between the light transmittance and the basis weight varies depending on the type of the sheet S, and thus it is required to select an optimum basis weight conversion formula for each type of the sheet S. Furthermore, it is also possible to calculate a density [g/m.sup.3] of the sheet S by dividing the basis weight [g/m.sup.3] by the thickness [m].

(91) In detecting the grain direction of the sheet S, as the media sensor, for example, an imaging device is used to photograph a transmission image of the sheet S being conveyed while receiving LED light applied thereto from a rear side thereof. The transmission image thus photographed is compared with a reference image stored in advance in the main body control portion **203**, and thus the grain direction of the sheet S is detected. Alternatively, as the media sensor, an ultrasonic measuring device can also be used to measure a reflection waveform from the sheet S receiving ultrasonic waves applied thereto. The reflection waveform thus measured is compared with a reference waveform stored in advance in the main body control portion **203**, and thus the grain direction of the sheet S is detected.

(92) In detecting the type of the sheet S, there is used a paper type determination algorithm in which the density, a surface property (specular reflection light, diffusion light), the basis weight, or the like of the sheet S is used to determine the type of the sheet S.

(93) Furthermore, as the media sensor, there is used a moisture meter that detects a moisture content of the sheet S, and thus the amount of ink on the sheet S can be detected.

(94) Furthermore, while the foregoing embodiment exemplarily uses an inkjet recording apparatus as the image forming apparatus **200**, a printer or a copy machine employing an electrophotographic scheme can also be used as the image forming apparatus **200**. In an inkjet recording scheme in which ink is ejected to the sheet S, the sheet S is more likely to be curled than in the electrophotographic scheme. The present disclosure, therefore, is particularly useful for the sheet post-processor **1** to which an inkjet recording apparatus is connected as the image forming apparatus **200**.

(95) Furthermore, while the foregoing embodiment exemplarily uses, as the sheet discharger of the present disclosure, the first sheet discharge portion **4** in the sheet post-processor **1** connected to the image forming apparatus **200**, the sheet discharger of the present disclosure is applicable also to a sheet discharge portion in the image forming apparatus **200** used without being connected to the sheet post-processor **1**.

(96) The present disclosure is usable in a sheet discharger that discharges a sheet, and a sheet post-processor, an image forming apparatus, and an image forming system that are provided with the sheet discharger.

Claims

1. A sheet discharger, comprising: a sheet discharge outlet through which a sheet is discharged; a discharge member that is disposed at the sheet discharge outlet and conveys the sheet in a discharge direction; a sheet discharge tray that is disposed downstream from the sheet discharge outlet with respect to the discharge direction and on which the sheet discharged through the sheet discharge outlet is loaded; a sheet hold-down member that is disposed below the sheet discharge outlet and is swingable between a hold-down position at which the sheet loaded on the sheet discharge tray is held down at an upstream part thereof in the discharge direction and a retracted position retracted upward from the hold-down position; a drive portion that drives the sheet hold-down member; and a control portion that controls the drive portion, wherein the control portion is capable of executing, every time the sheet is discharged on the sheet discharge tray, a sheet hold-down operation in which

the sheet hold-down member disposed at the hold-down position is moved to the retracted position at a timing when a trailing end of the sheet discharged through the sheet discharge outlet passes through the discharge member, and the sheet hold-down member is moved from the retracted position to the hold-down position at a timing when loading of the sheet on the sheet discharge tray is completed, and the control portion determines, in accordance with characteristics of the sheet, whether or not to execute the sheet hold-down operation, based on output information related to the characteristics of the sheet, the control portion estimates a direction of curl of the sheet discharged through the sheet discharge outlet, and when the sheet discharged through the sheet discharge outlet is uncurled or curled upwardly, the sheet hold-down operation is executed, while when the sheet is curled downwardly, the sheet hold-down member is maintained to be disposed at the retracted position.

2. The sheet discharger according to claim 1, wherein the output information related to the characteristics of the sheet includes at least one of a thickness of the sheet, a basis weight of the sheet, a grain direction of the sheet, and an orientation of a printing surface of the sheet.
 3. A sheet post-processor, comprising: a processing section that performs a prescribed type of post-processing on a sheet; and the sheet discharger according to claim 1, the sheet discharger discharging the sheet subjected to the prescribed type of post-processing by the processing section.
 4. An image forming system, comprising: an image forming apparatus that forms an image on a sheet; and the sheet post-processor according to claim 3, the sheet post-processor performing the prescribed type of post-processing on the sheet on which the image has been formed by the image forming apparatus.
 5. An image forming system, comprising: an image forming apparatus that forms an image on a sheet; and a sheet post-processor including: a processing section that performs a prescribed type of post-processing on the sheet; and the sheet discharger according to claim 1, the sheet discharger discharging the sheet subjected to the prescribed type of post-processing by the processing section, wherein the image forming apparatus is an inkjet recording apparatus that ejects ink to the sheet so as to perform image recording thereon, and the output information related to the characteristics of the sheet includes an amount of ink used for the image recording on the sheet.
 6. An image forming apparatus, comprising: an image forming portion that forms an image on a sheet; and the sheet discharger according to claim 1, the sheet discharger discharging the sheet on which the image has been formed by the image forming portion.
 7. The image forming apparatus according to claim 6, wherein the image forming portion ejects ink to the sheet so as to perform image recording thereon.
 8. An image forming system, comprising: an image forming apparatus that forms an image on a sheet; and a sheet post-processor including: a processing section that performs a prescribed type of post-processing on the sheet; and the sheet discharger according to claim 2, the sheet discharger discharging the sheet subjected to the prescribed type of post-processing by the processing section, wherein the image forming system further includes: an input portion for inputting the output information related to the characteristics of the sheet.
 9. The image forming system according to claim 8, wherein a position of the sheet hold-down member is changeable through an input operation with respect to the input portion.
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